

PKCERT AI & Software Development Internship

Task 15 – Model Persistence & Mini-Project

Objective:

Develop practical skills in saving and loading trained machine learning models using Pickle and Joblib, and apply a complete end-to-end ML pipeline (EDA → Preprocessing → Model → Evaluation) on a new dataset as part of a mini-project.

Total Marks: 100

Part A – Saving Trained Models (20 Marks)

1. Train any classification or regression model on a dataset of your choice.
2. Save the trained model using the pickle module.
3. Save the same trained model using joblib.

Part B – Loading & Verifying Models (20 Marks)

1. Load the saved model back into memory using both pickle and joblib.
2. Verify that the loaded model produces the same predictions as the original trained model.
3. Explain the differences between pickle and joblib, and when to use each.

Part C – End-to-End ML Pipeline (45 Marks)

1. Select a new public dataset (different from datasets used in previous tasks).
2. Perform Exploratory Data Analysis (EDA) to understand the dataset's structure and patterns.
3. Preprocess the dataset, including handling missing values, encoding, and scaling as needed.
4. Train an appropriate machine learning model on the processed data.
5. Evaluate the model using suitable performance metrics.

Part D – Model Saving & Documentation (15 Marks)

1. Save the final trained model from the mini-project using pickle or joblib.
2. Summarize the pipeline, key findings, and challenges faced during the mini-project.

Assessment Area	Marks
Saving Trained Models	20
Loading & Verifying Models	20
End-to-End ML Pipeline (Mini-Project)	45
Model Saving & Documentation	15
Total	100